Course Code: 23CSH01T

R23

PRAGATI ENGINEERING COLLEGE: SURAMPALEM
(AUTONOMOUS)

II B.Tech II Semester Regular Examinations, May - 2025

PYTHON FOR DATA SCIENCE

CSE(Honors)

Time: 3 hours | **Max. Marks: 70**

Note:

i. *Question No. 1 shall contain 10 compulsory short answer questions (2 questions from each unit) for a total of 20 marks such that each question carries 2 marks.*

ii. *In each of the questions from 2 to the last question, there shall be either/or type questions of 10 marks each.*

Q. No.		Questions	BTL	CO	Marks
1.	a)	What is datafication?	K1	CO1	2M
	b)	What are the steps in the data science process?	K1	CO1	2M
	c)	What is the purpose of the drop() function in pandas?	K2	CO2	2M
	d)	Define reindexing in pandas.	K1	CO2	2M
	e)	What is the purpose of read_csv() and to_csv() in pandas?	K2	CO3	2M
	f)	What function in pandas is used to read XML data into a DataFrame?	K1	CO3	2M
	g)	What is the role of SQLite in handling text mining and analytics?	K1	CO4	2M
	h)	Define Cypher.	K1	CO4	2M
	i)	List out some popular tools for developing interactive dashboards?	K2	CO5	2M
	j)	What is cross-filtering in the context of data visualization?	K2	CO5	2M
		UNIT – I			
2.	a)	What is Exploratory Data Analysis (EDA)? Describe its importance and key techniques.	K2	CO1	5M
	b)	What is an ndarray in NumPy? Explain its features and how it is different from a Python list.	K2	CO1	5M
		OR			
3.	a)	Explain how to sort and retrieve unique elements from a NumPy array. Write code to demonstrate both operations.	K3	CO1	10M
	b)	What is Boolean Indexing in NumPy? How is it useful in data filtering?	K2	CO1	5M
		UNIT – II			
4.	a)	Explain the difference between Series and DataFrame in pandas. Give one use case for each.	K3	CO2	5M
	b)	Discuss how pandas handles missing data. Explain methods to detect, filter, and fill missing values.	K2	CO2	5M
		OR			
5.	a)	Write code to drop specific rows and columns from a DataFrame. Explain the use of the axis parameter in this context	K3	CO2	5M

	b)	What is the use of unique() and value_counts() in pandas? How do they help in data analysis?	K2	CO2	5M
		UNIT – III			
6.	a)	What is MongoDB? Briefly explain how data is stored and retrieved in MongoDB.	K2	CO3	5M
	b)	Compare JSON, XML, and HTML formats. How is each used in data analysis?	K2	CO3	5M
		OR			
7.	a)	What are delimited formats? How can one manually handle them when automated parsing fails?	K2	CO3	5M
	b)	What is the HDF5 format? Describe its features and advantages for storing large datasets.	K2	CO3	5M
		UNIT – IV			
8.	a)	List five applications of graph databases in real-world scenarios	K1	CO4	5M
	b)	What is the purpose of the NLTK library in Python? List four tasks it can perform.	K1	CO4	5M
		OR			
9.	a)	List the major steps in a basic text classification process using Python.	K1	CO4	5M
	b)	What is text mining? List three examples of text mining applications.	K1	CO4	5M
		UNIT – V			
10.		Explain the process of integrating different types of data visualizations (e.g., bar charts, pie charts, and line graphs) into a single dashboard.	K2	CO5	10M
		OR			
11.	a)	Describe the role of JavaScript libraries in the creation of interactive and real-time data dashboards.	K2	CO5	5M
	b)	Explain in detail different types of cross-filters.	K2	CO5	5M